

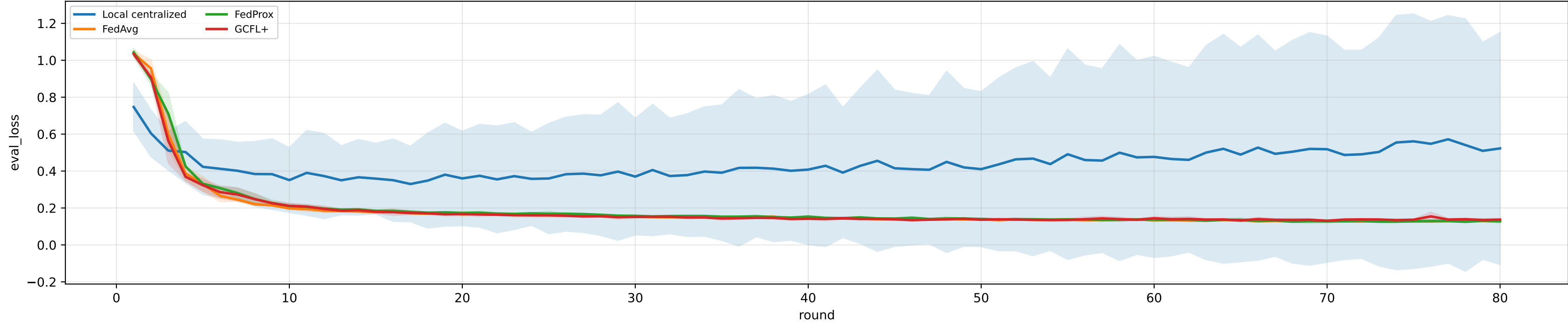
Controlled task-specialization level: High | aggregated / pooled performance over 5 clients
| model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% | cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 | family_centroid_jsd_mean=0.161

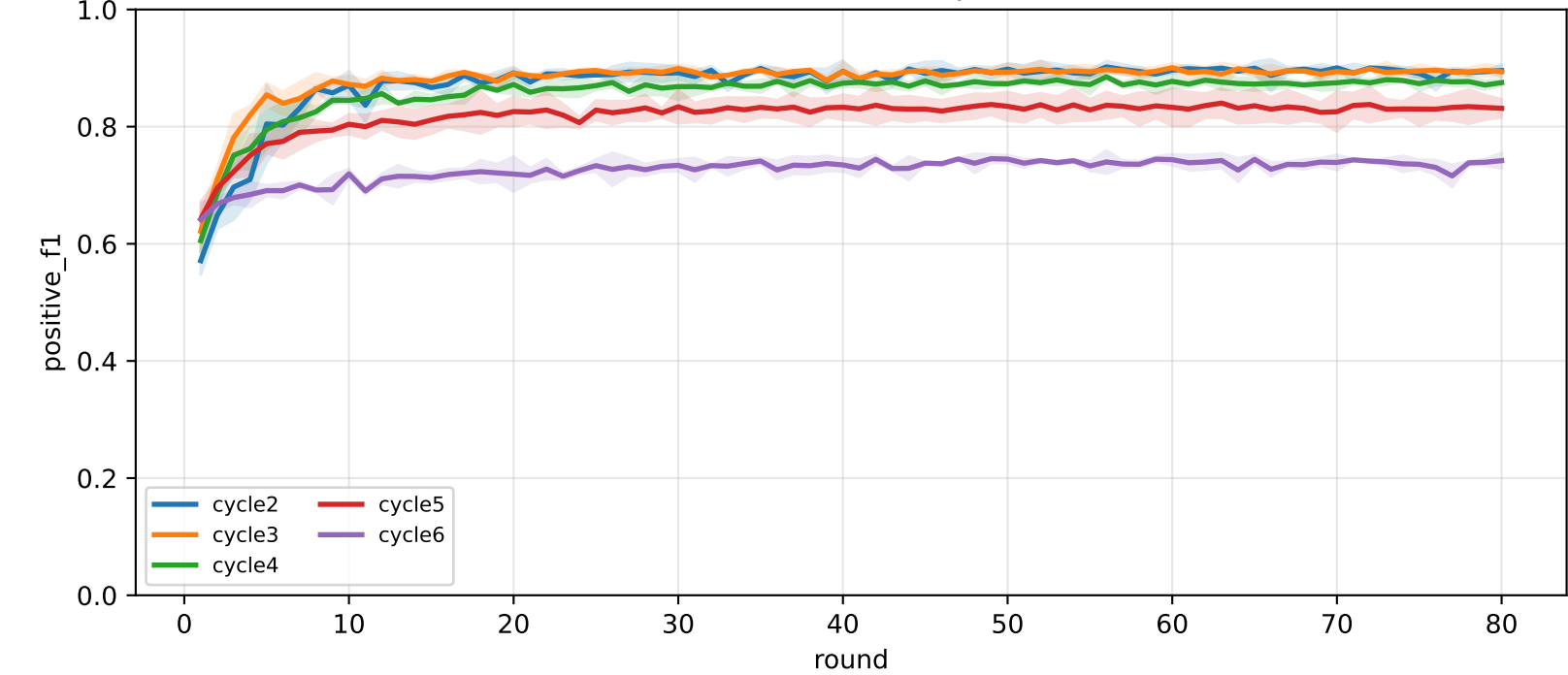
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	84.1% ± 0.3%	85.3% ± 0.3%	87.0% ± 1.6%	88.2% ± 0.8%	86.8% ± 0.5%	83.1% ± 1.1%	73.7% ± 0.3%
FedAvg	92.5% ± 0.4%	93.3% ± 0.3%	98.8% ± 0.1%	97.5% ± 0.4%	94.5% ± 0.7%	91.5% ± 1.4%	84.2% ± 1.0%
FedProx	92.8% ± 0.4%	93.6% ± 0.3%	98.9% ± 0.2%	97.9% ± 0.3%	95.3% ± 0.7%	92.4% ± 1.9%	83.7% ± 0.8%
GCFL+	92.3% ± 0.2%	93.2% ± 0.2%	98.7% ± 0.1%	97.9% ± 0.2%	94.3% ± 1.4%	91.8% ± 1.0%	83.2% ± 1.9%

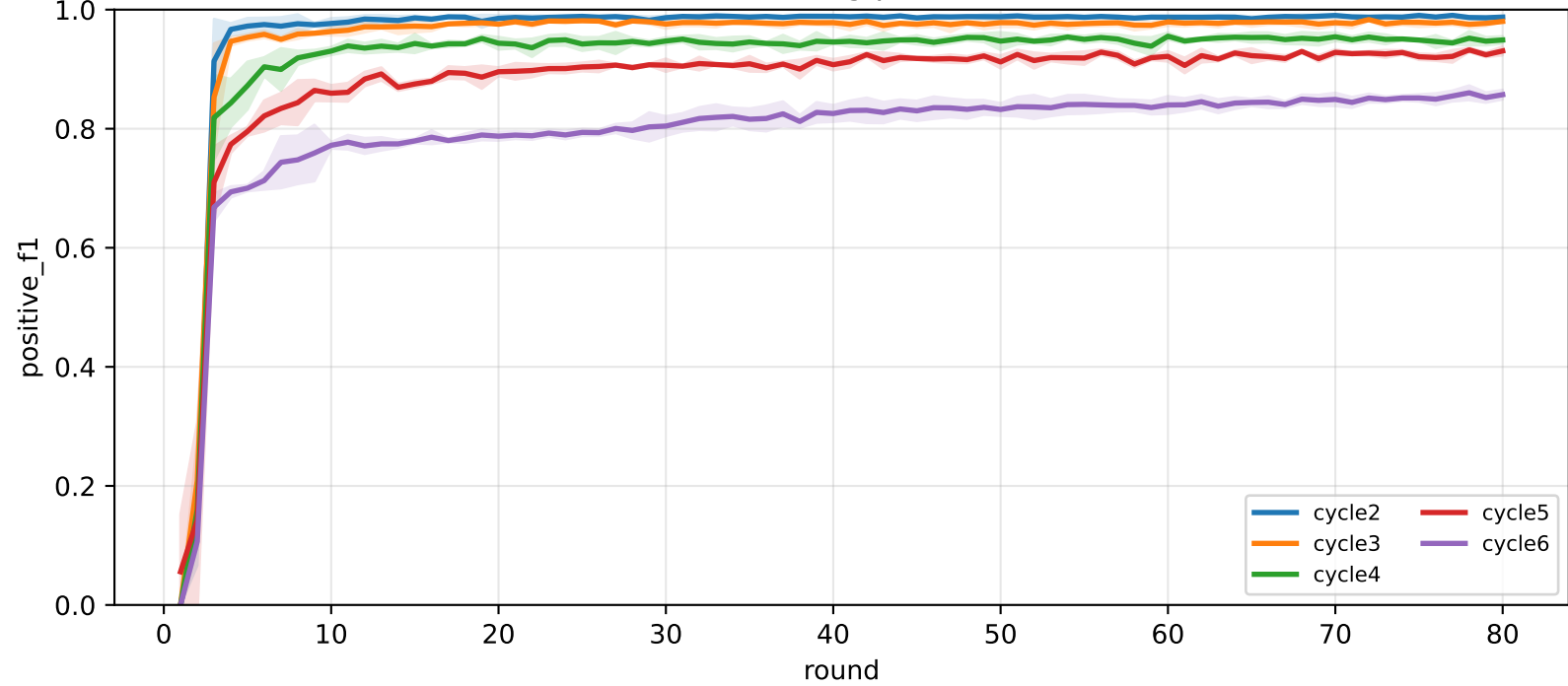
Validation loss comparison



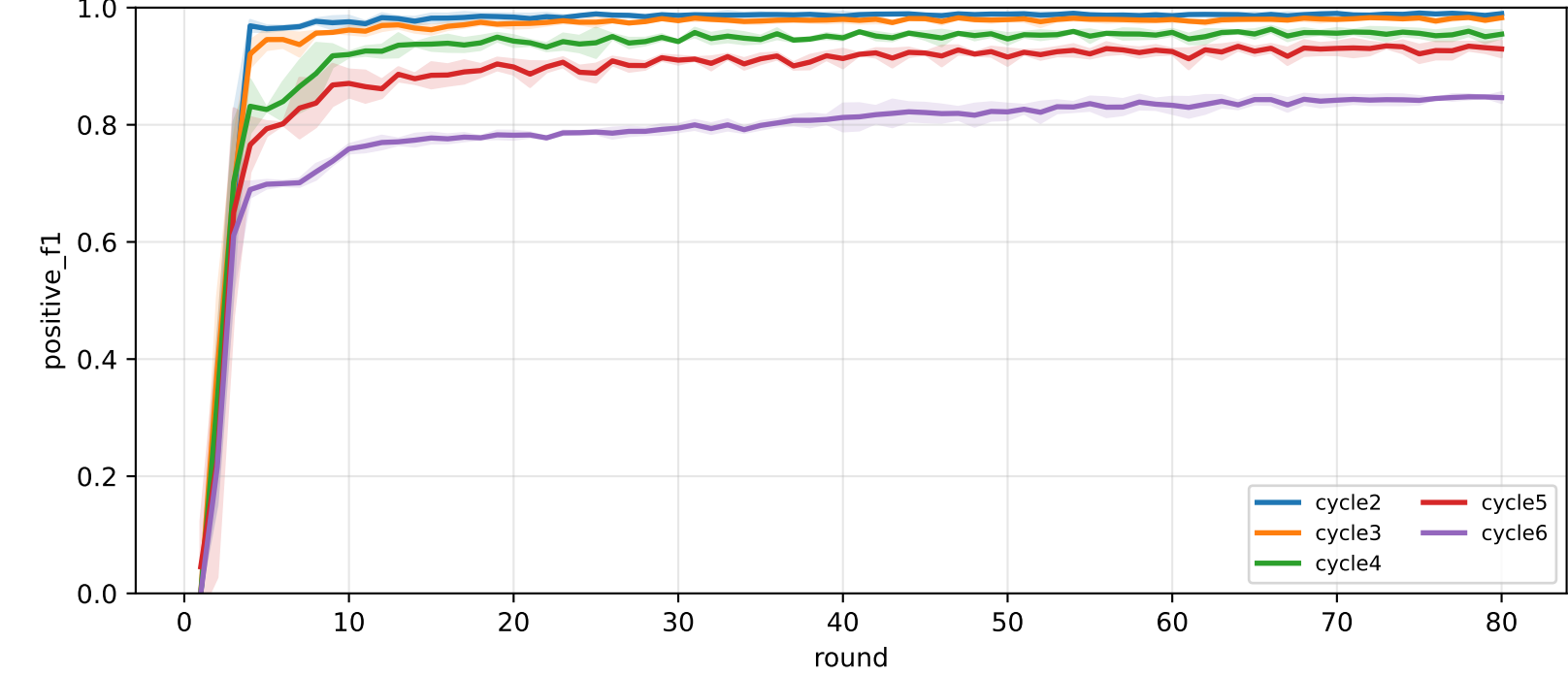
Pooled local centralized positive F1



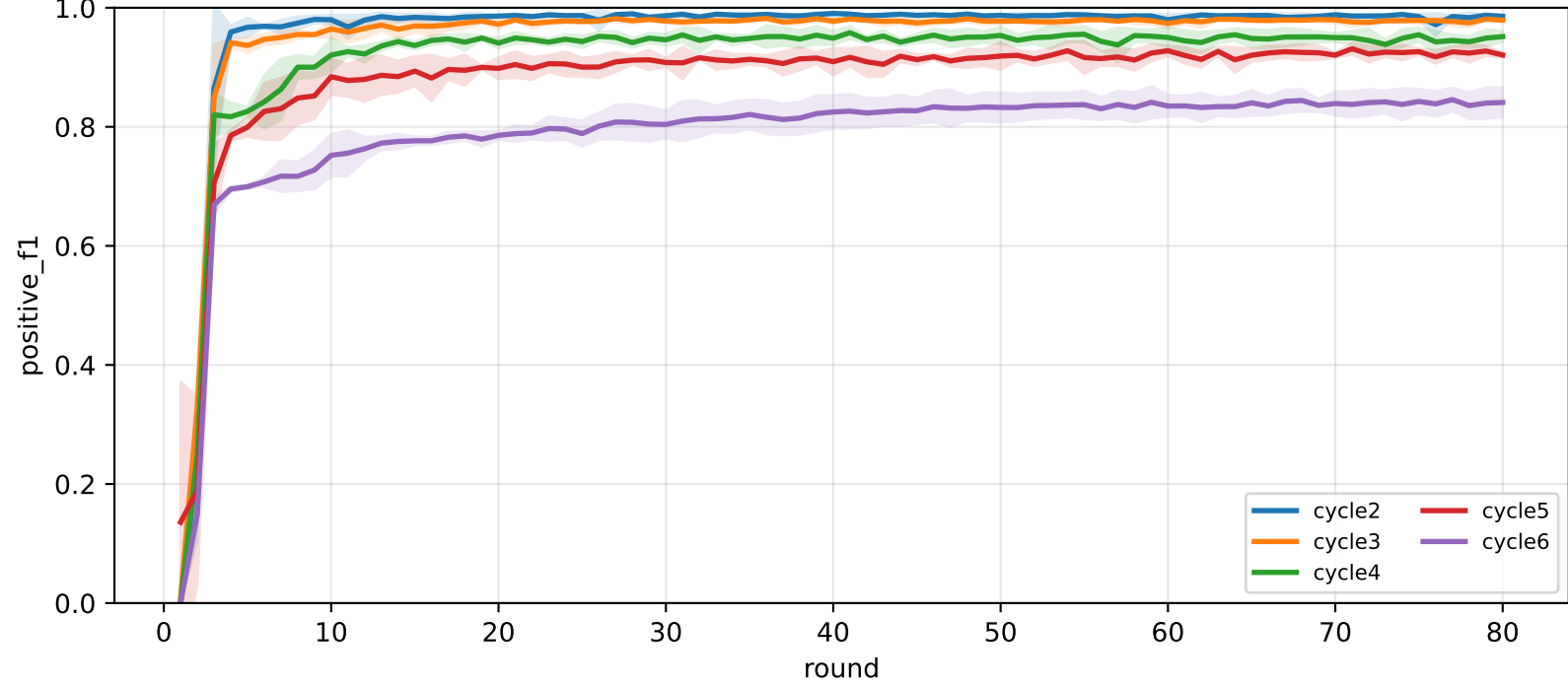
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



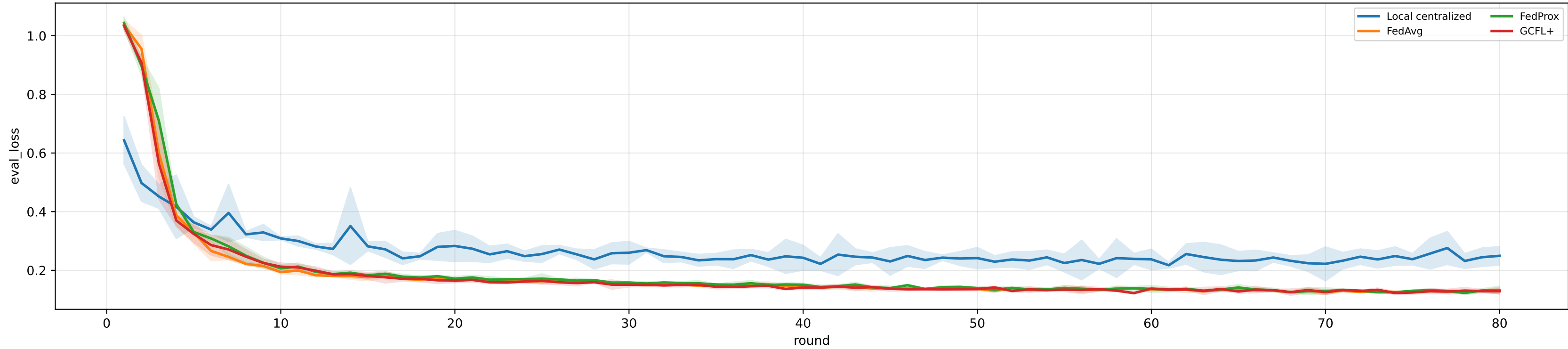
Controlled task-specialization level: High | family aggregated / pooled performance |
family=specialized_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 19.4%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% |
cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 |
family_centroid_jsd_mean=0.161

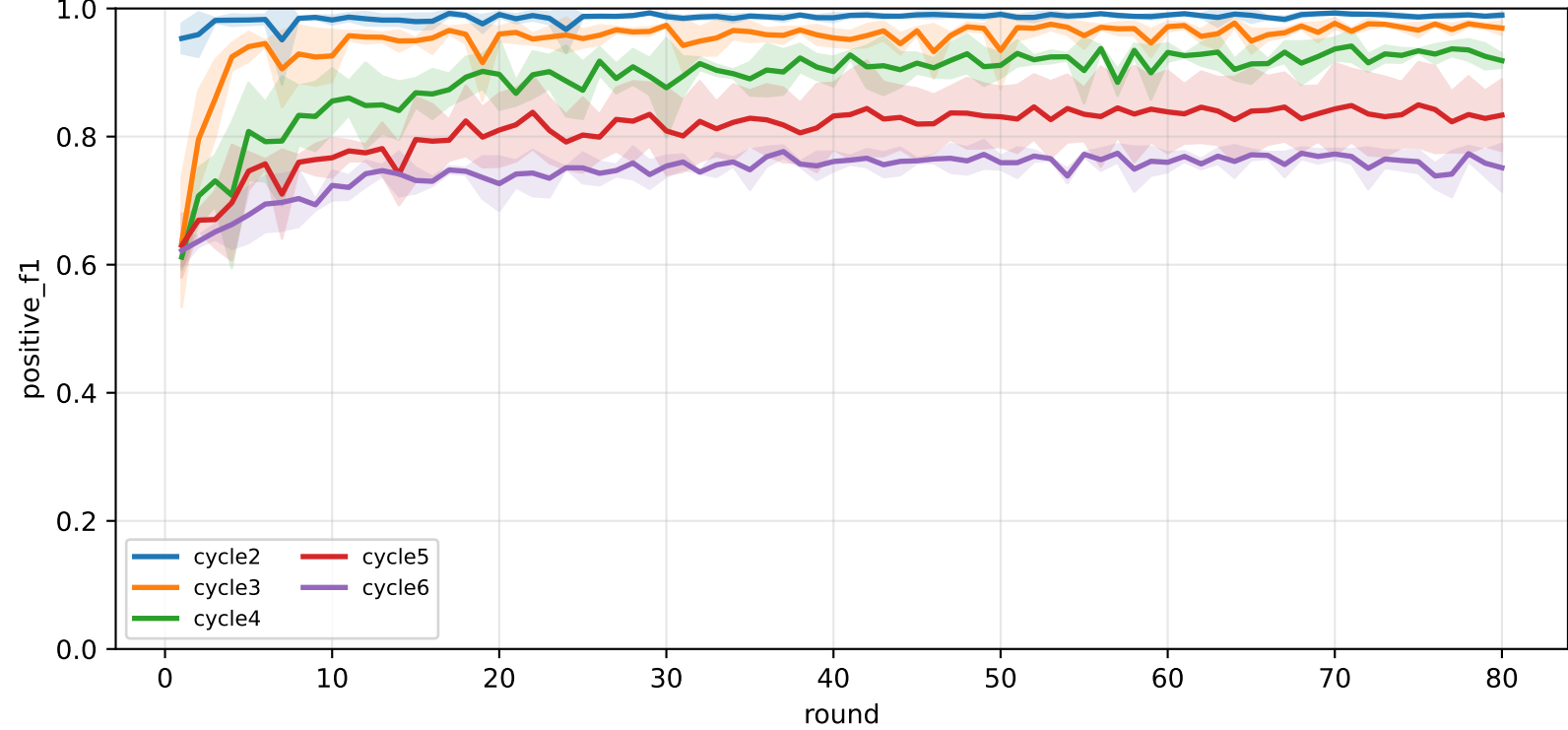
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	88.3% ± 1.5%	89.8% ± 1.2%	99.0% ± 0.2%	97.2% ± 0.4%	93.2% ± 0.6%	83.8% ± 5.3%	75.6% ± 1.0%
FedAvg	92.7% ± 0.6%	93.5% ± 0.4%	99.0% ± 0.4%	97.7% ± 0.3%	94.9% ± 0.8%	91.8% ± 1.5%	84.2% ± 1.3%
FedProx	93.0% ± 0.4%	93.8% ± 0.3%	99.1% ± 0.4%	98.0% ± 0.5%	95.4% ± 0.6%	92.5% ± 2.1%	83.9% ± 1.0%
GCFL+	92.5% ± 0.1%	93.3% ± 0.1%	98.8% ± 0.2%	98.2% ± 0.2%	93.9% ± 1.8%	92.0% ± 1.2%	83.8% ± 1.6%

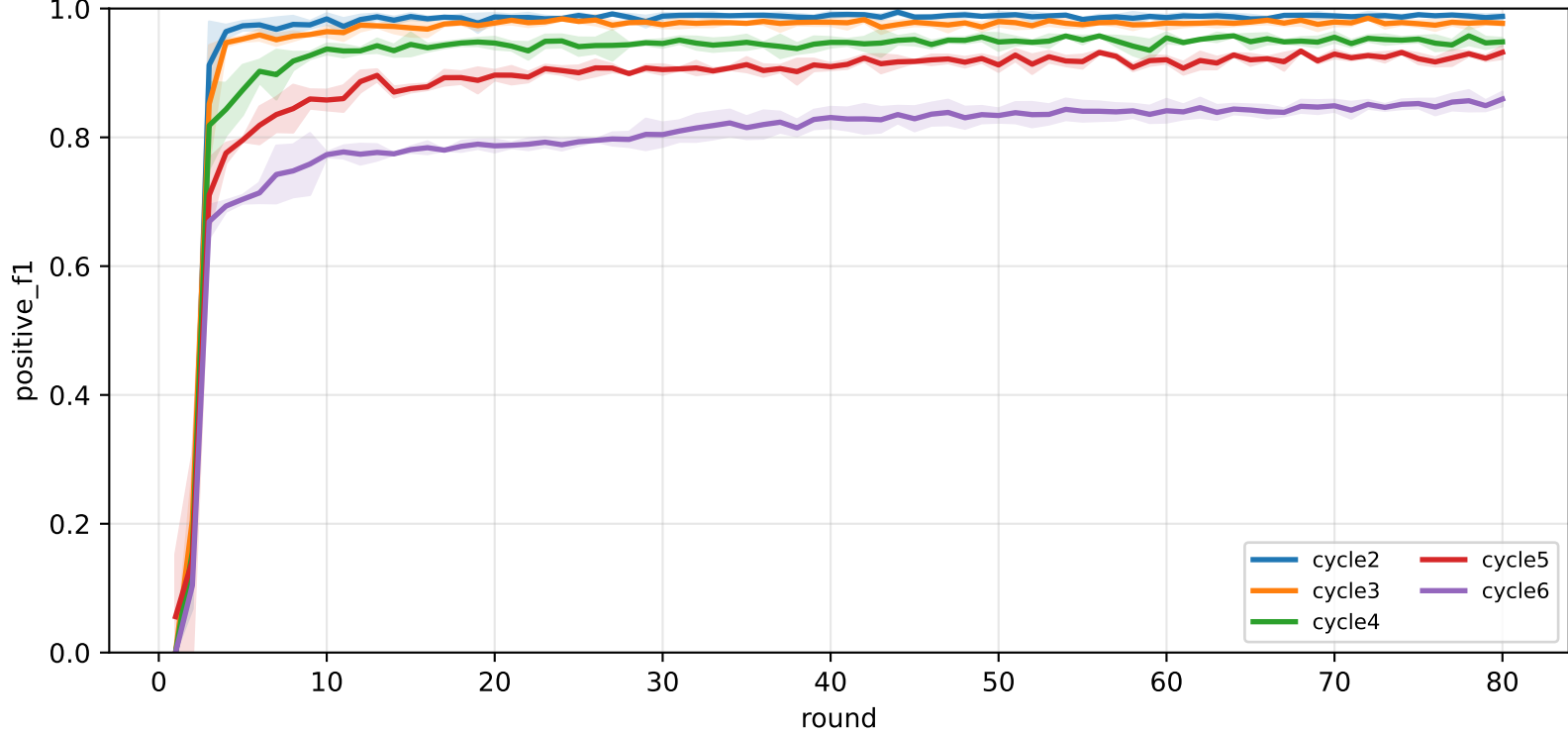
Validation loss comparison



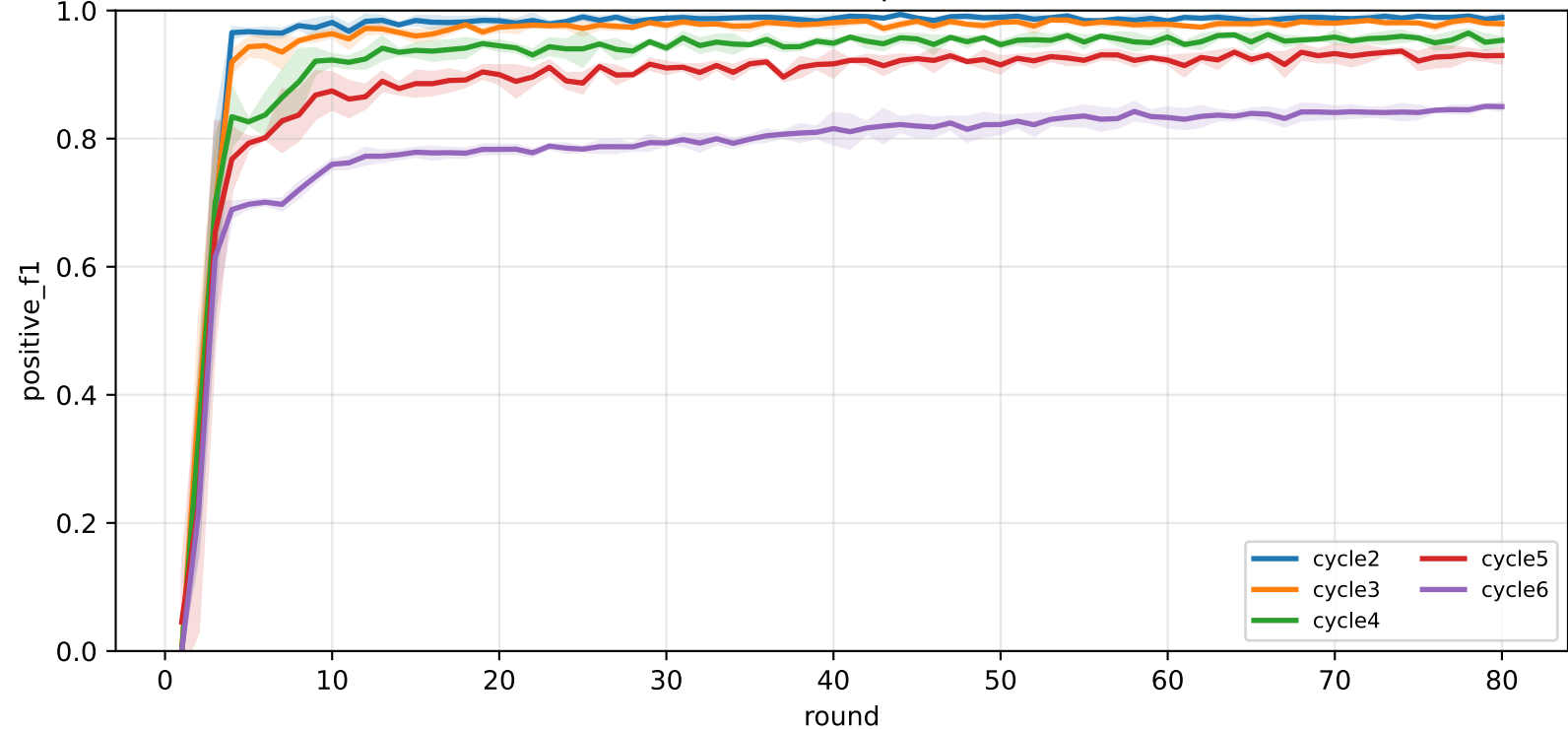
Pooled local centralized positive F1



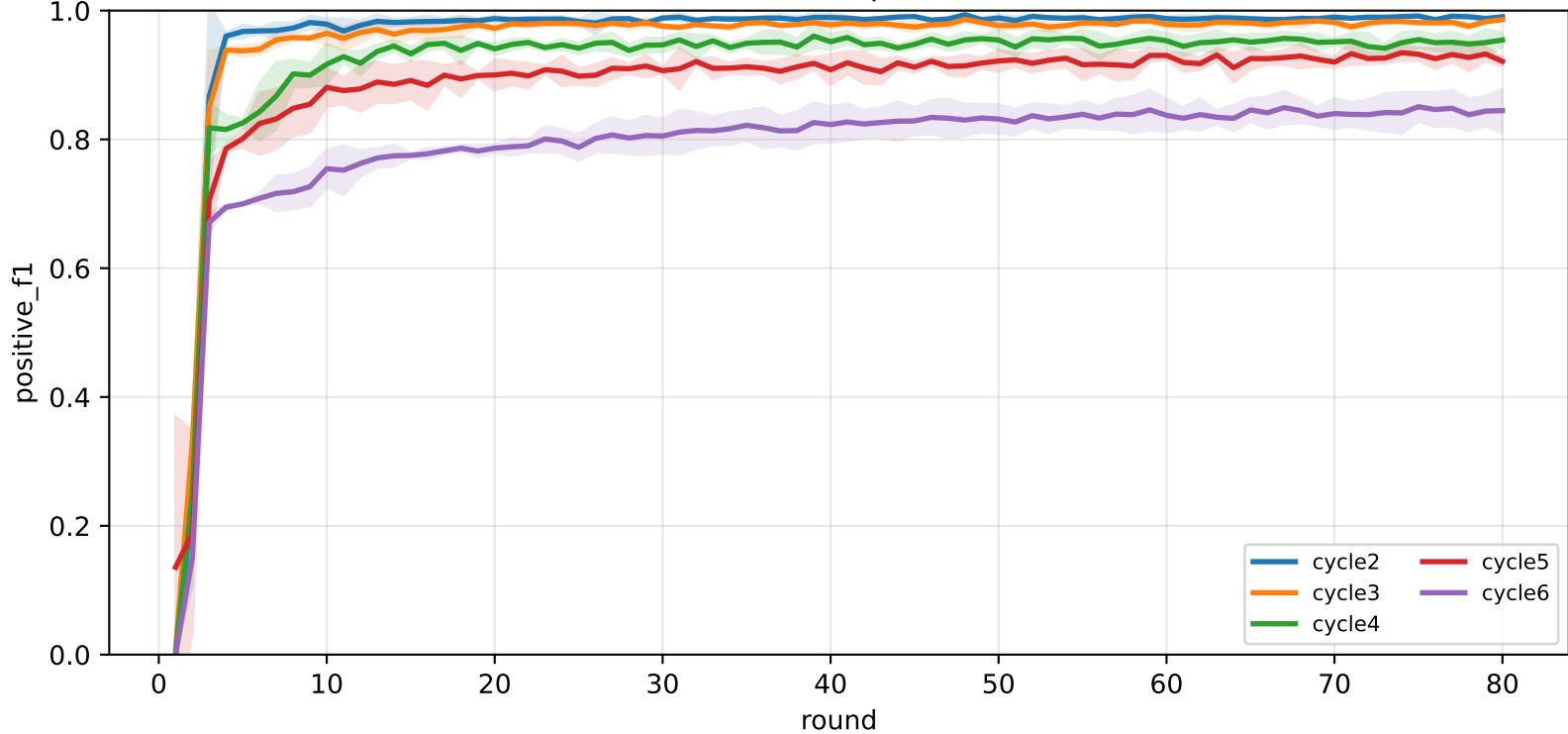
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



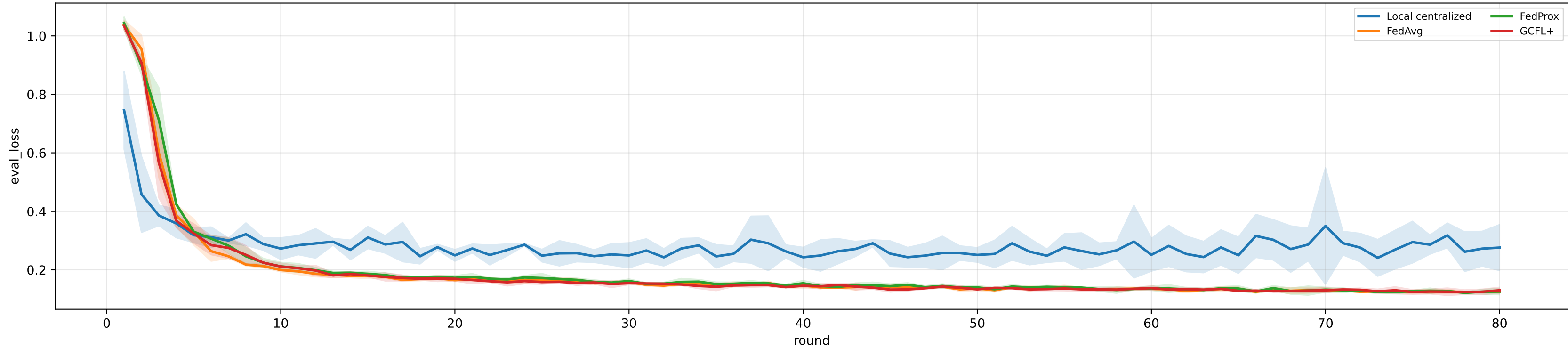
Controlled task-specialization level: High | family aggregated / pooled performance |
family=specialized_cycle3 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle3 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 23.6%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% |
cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 |
family_centroid_jsd_mean=0.161

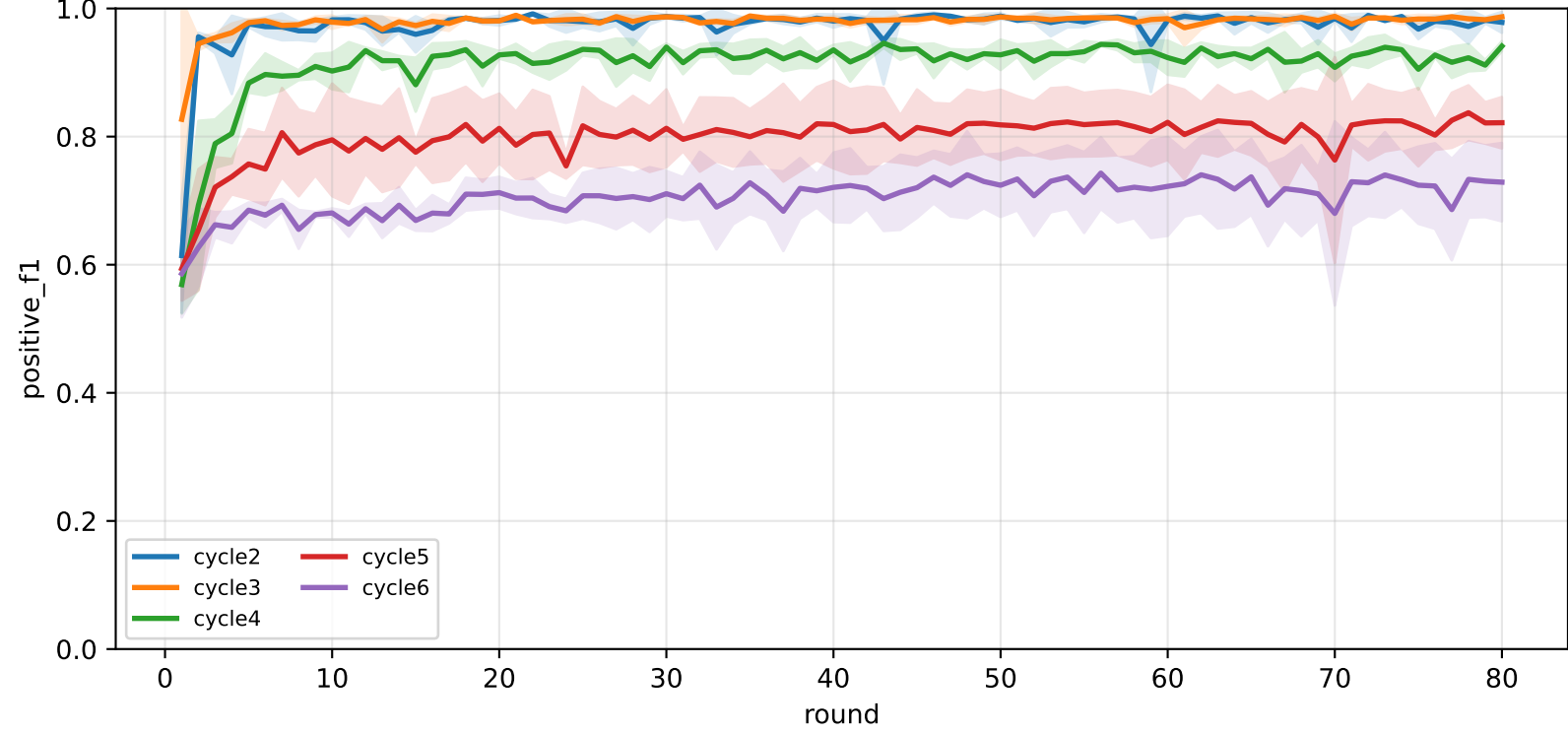
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	87.2% ± 2.5%	89.2% ± 1.7%	98.4% ± 0.6%	98.4% ± 0.3%	94.1% ± 0.6%	82.5% ± 4.4%	72.3% ± 4.5%
FedAvg	92.3% ± 0.2%	93.1% ± 0.2%	98.7% ± 0.3%	97.1% ± 0.4%	94.4% ± 0.9%	91.4% ± 1.2%	84.1% ± 1.1%
FedProx	92.6% ± 0.3%	93.4% ± 0.3%	98.8% ± 0.3%	97.5% ± 0.2%	95.1% ± 1.1%	92.3% ± 1.7%	83.6% ± 1.1%
GCFL+	92.4% ± 0.2%	93.2% ± 0.1%	98.7% ± 0.4%	98.1% ± 0.1%	93.8% ± 1.6%	92.0% ± 1.1%	83.5% ± 1.8%

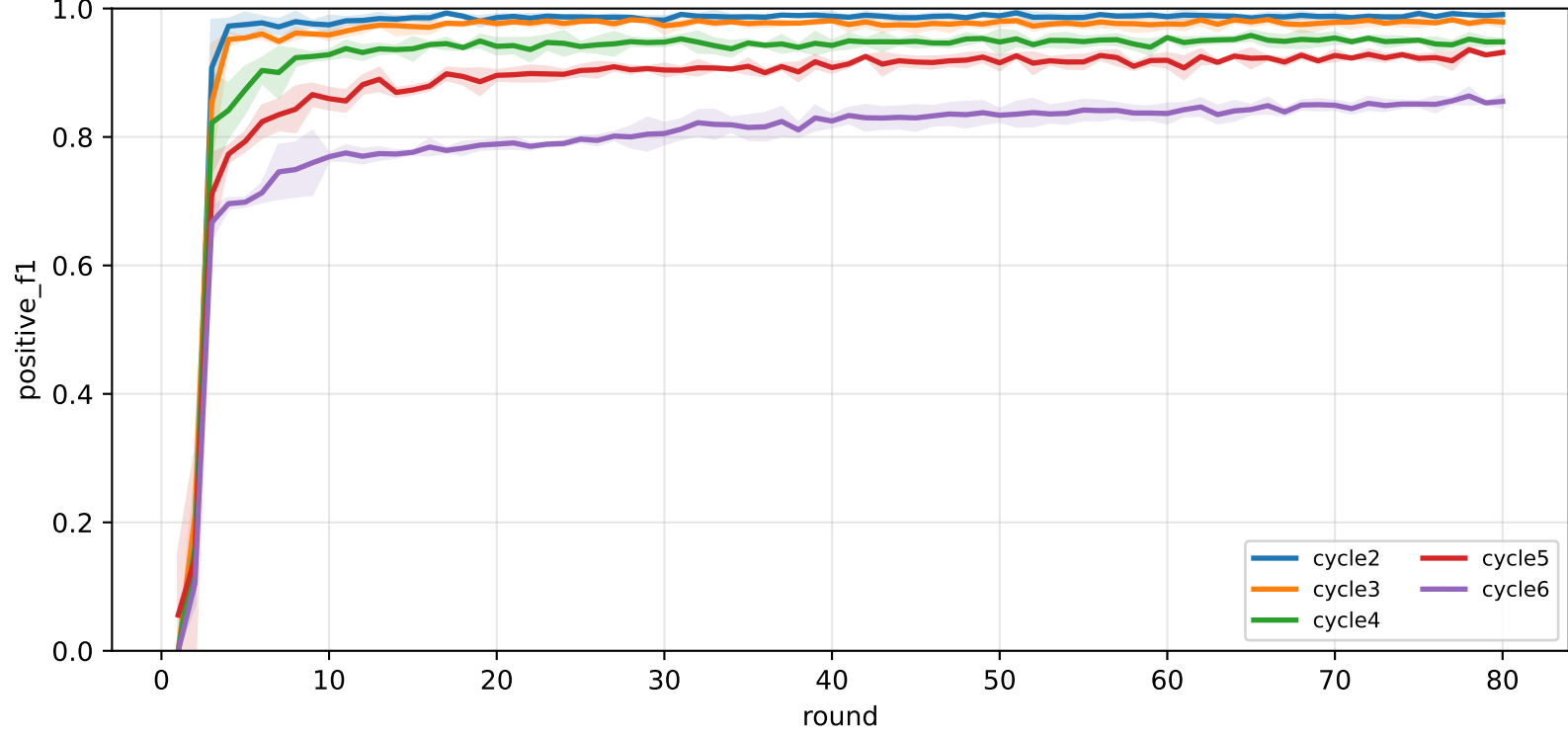
Validation loss comparison



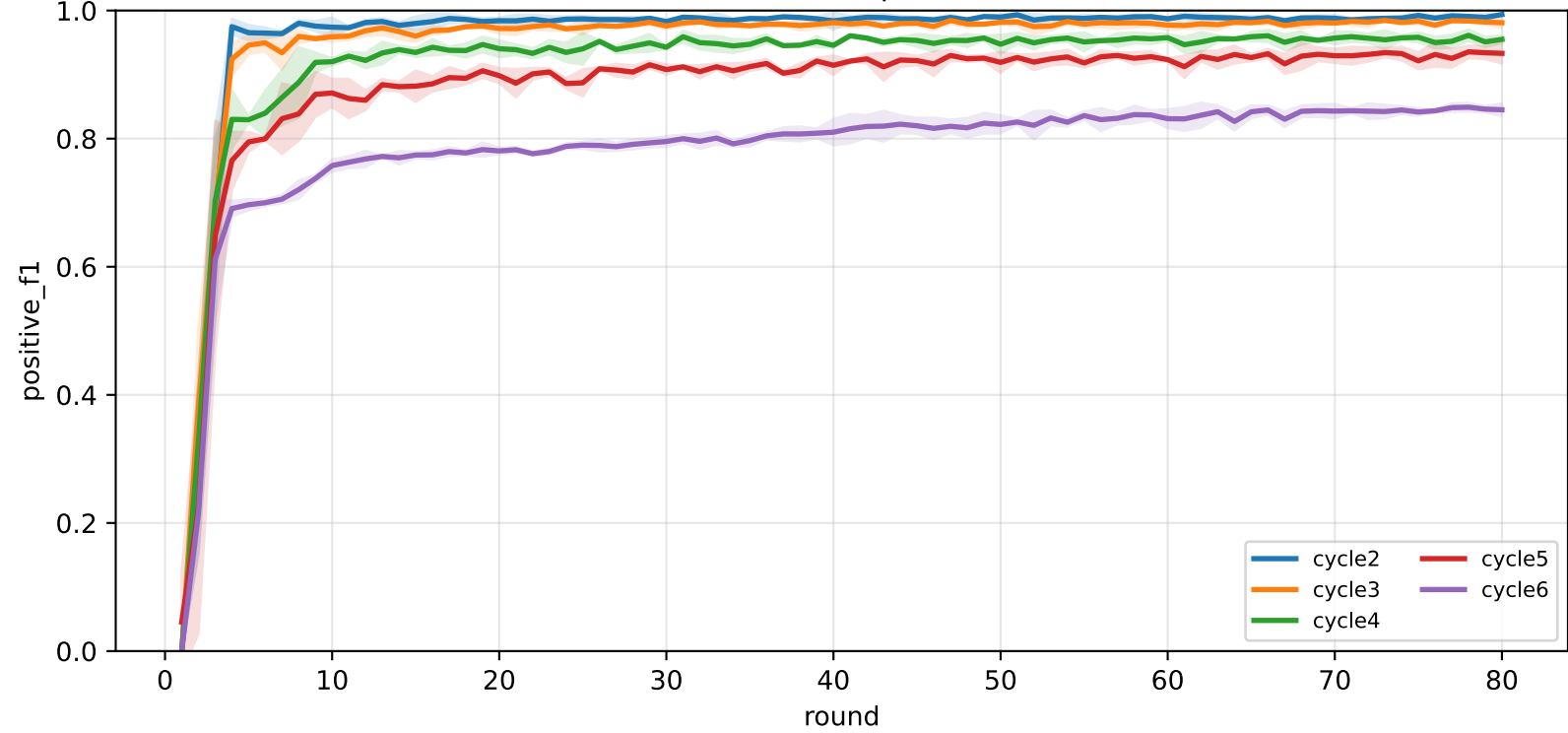
Pooled local centralized positive F1



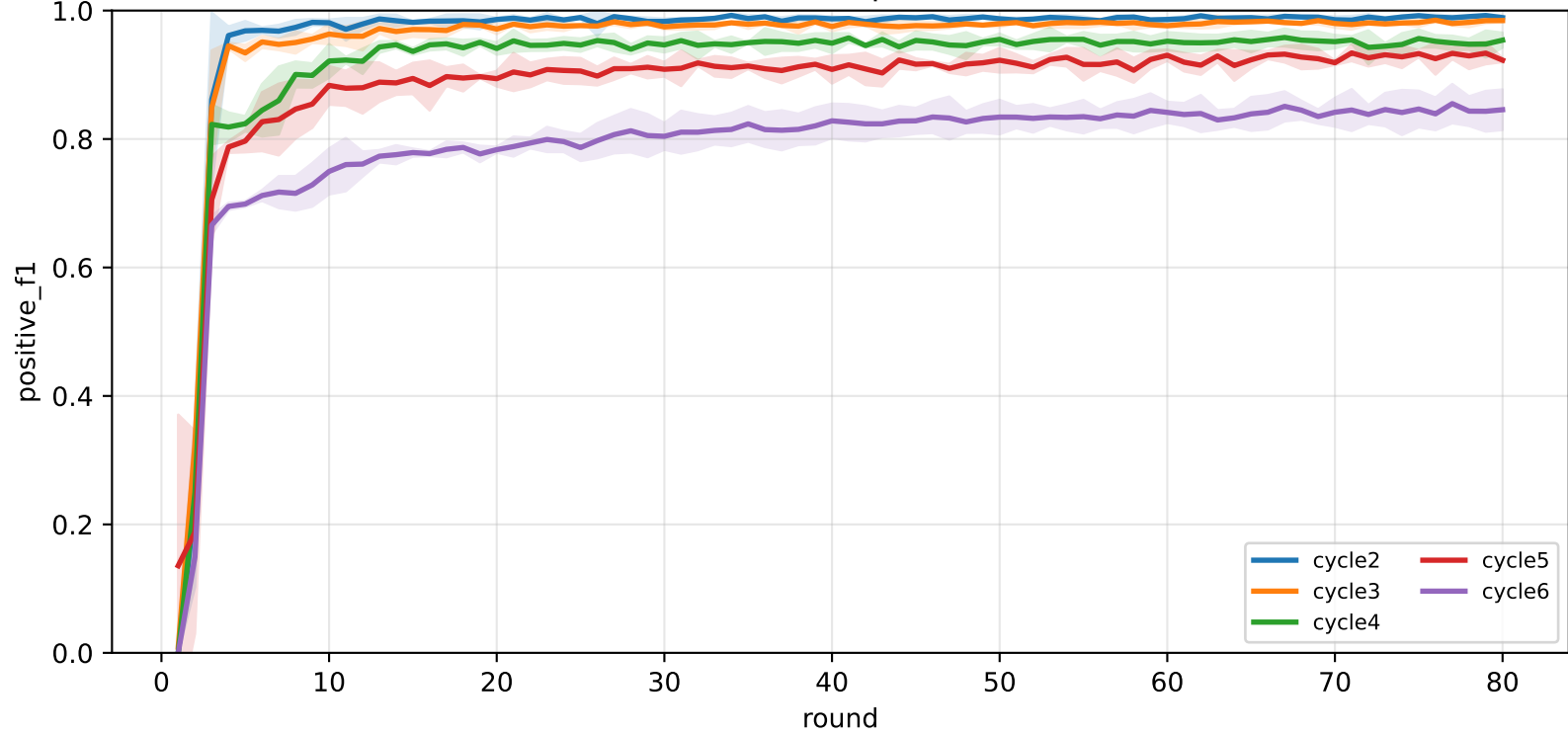
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



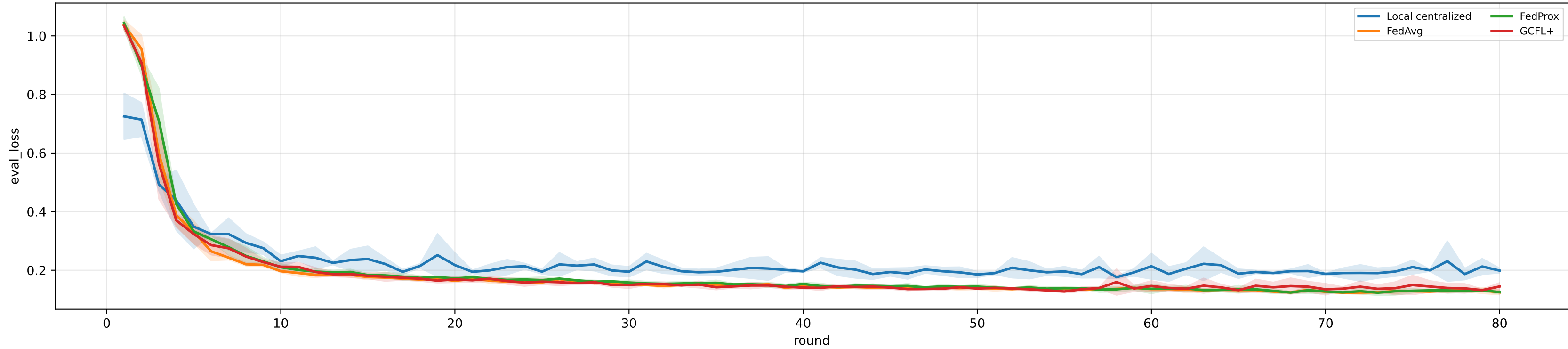
Controlled task-specialization level: High | family aggregated / pooled performance |
family=specialized_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle4 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 25.6%±0.0% / true 25.6% |
cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 |
family_centroid_jsd_mean=0.161

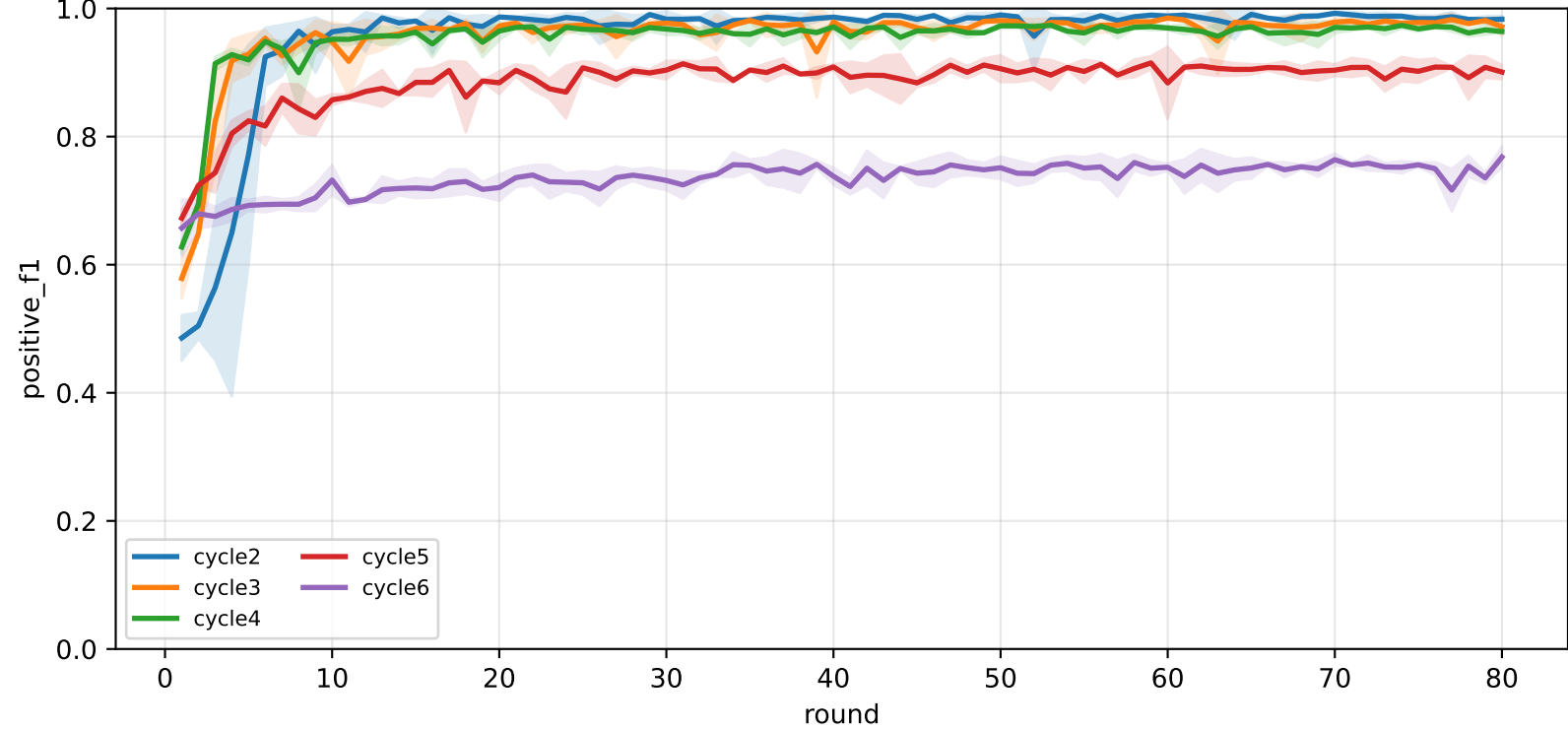
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	89.8% ± 0.2%	91.4% ± 0.1%	99.0% ± 0.3%	97.8% ± 0.2%	97.0% ± 0.3%	89.3% ± 0.6%	74.0% ± 0.3%
FedAvg	92.4% ± 0.4%	93.3% ± 0.3%	98.9% ± 0.1%	97.3% ± 0.4%	94.6% ± 0.7%	91.2% ± 1.4%	84.2% ± 0.9%
FedProx	92.7% ± 0.4%	93.6% ± 0.3%	99.0% ± 0.2%	97.6% ± 0.2%	95.3% ± 0.5%	92.0% ± 2.0%	83.8% ± 0.5%
GCFL+	92.2% ± 0.0%	93.1% ± 0.1%	98.7% ± 0.6%	97.7% ± 0.3%	95.3% ± 0.6%	91.4% ± 1.2%	82.5% ± 1.5%

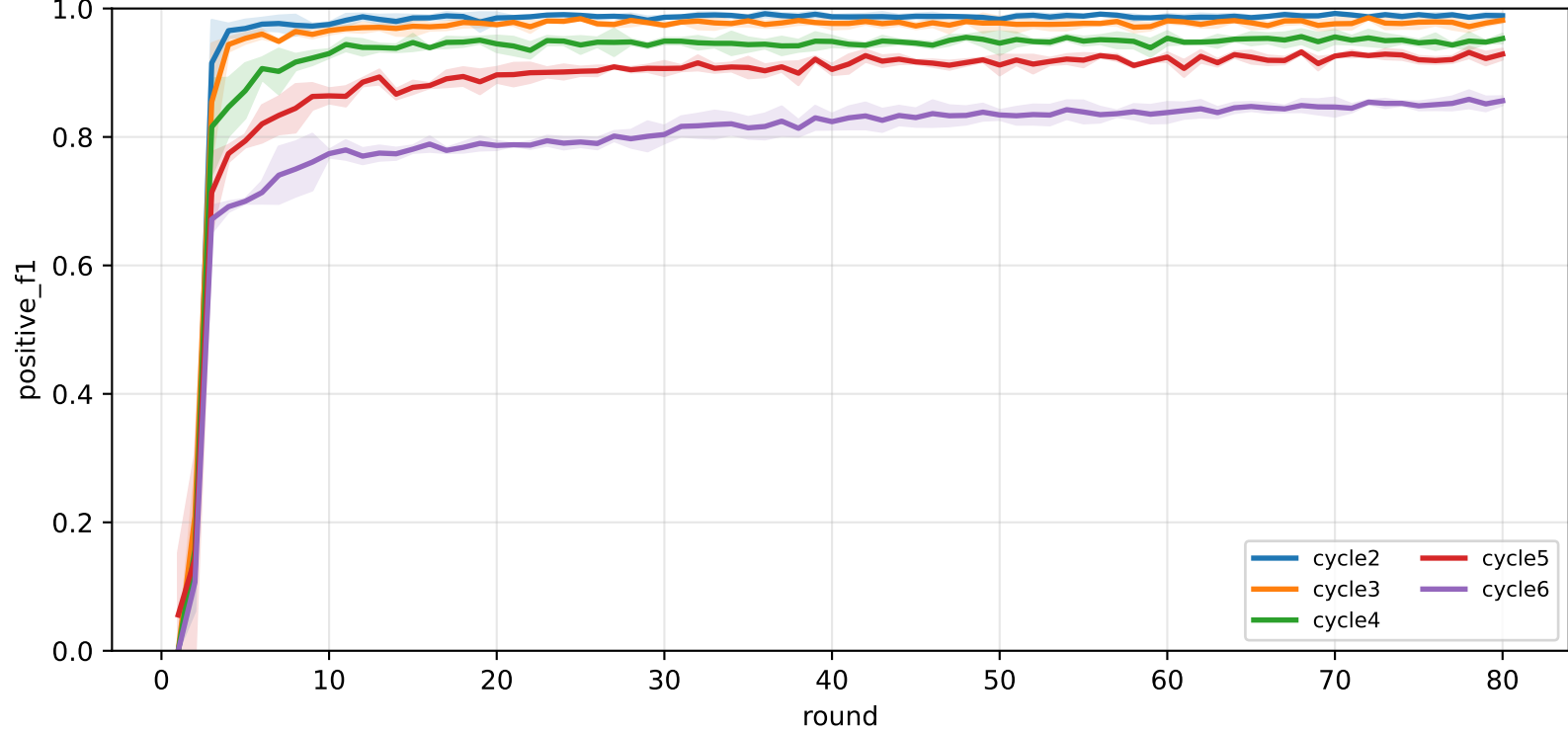
Validation loss comparison



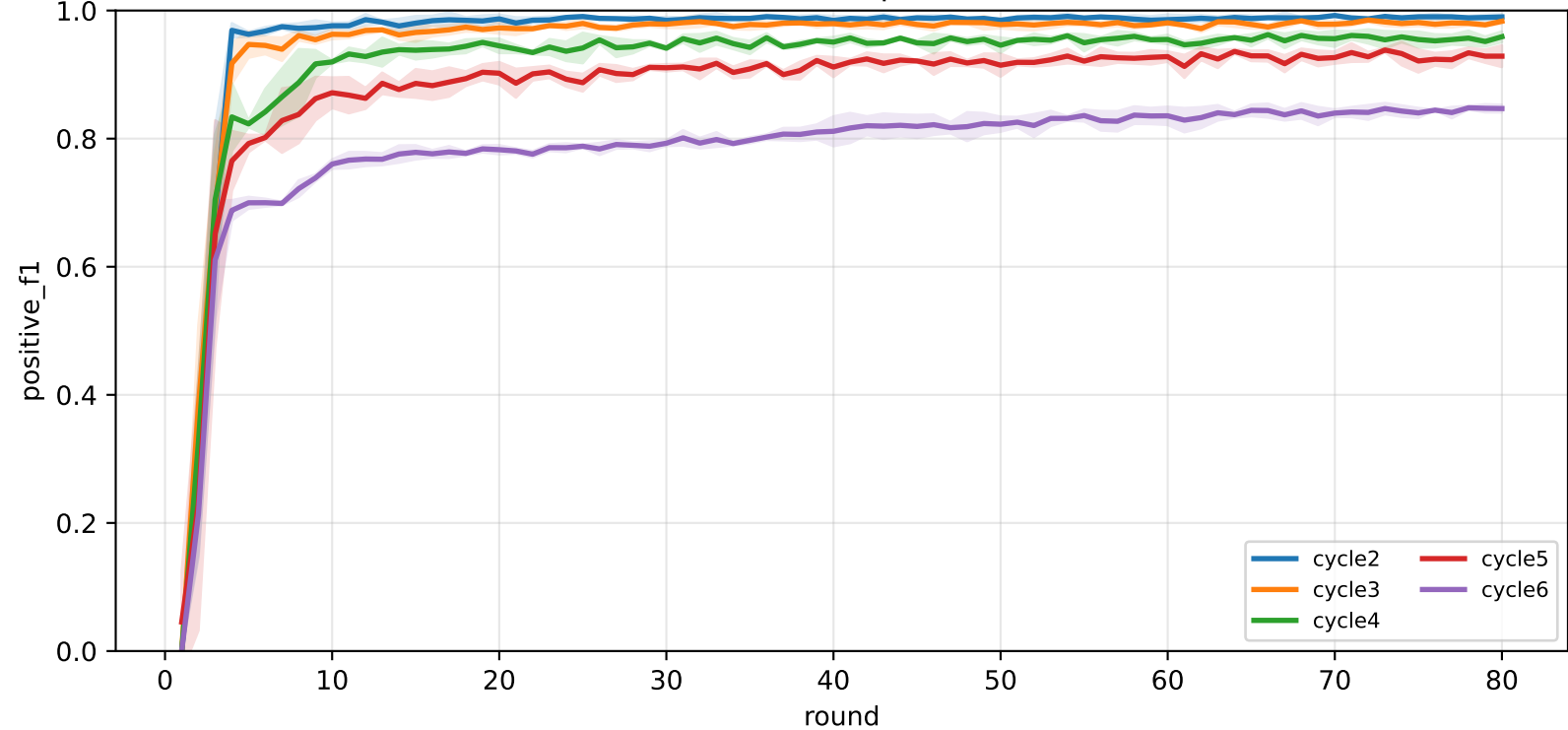
Pooled local centralized positive F1



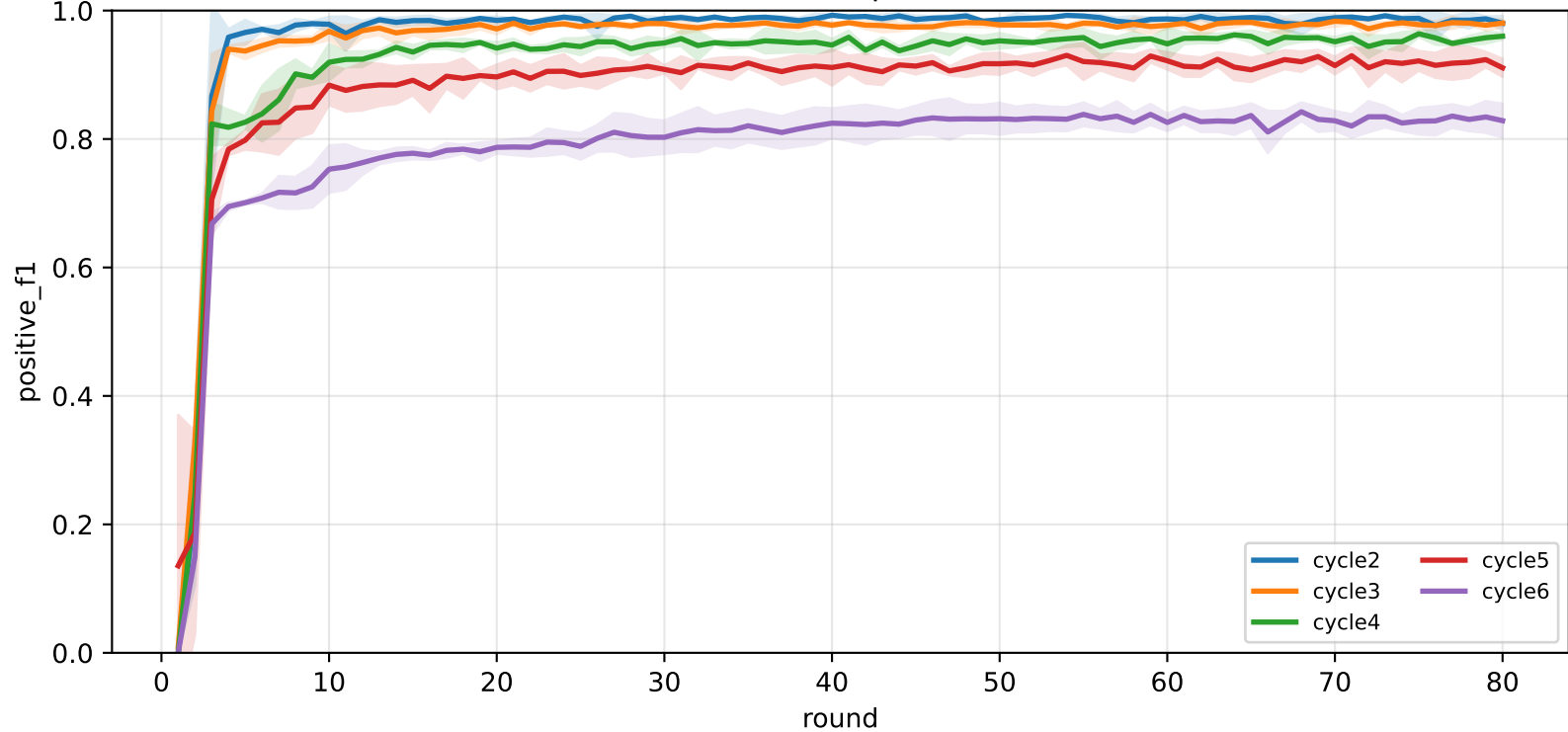
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



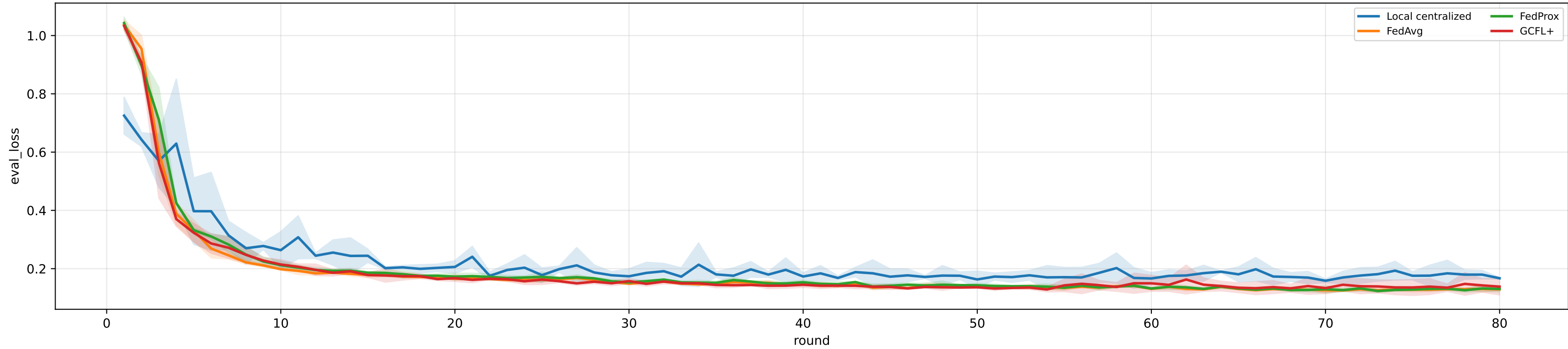
Controlled task-specialization level: High | family aggregated / pooled performance |
family=specialized_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle5 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% |
cycle5=visible 25.7%±0.0% / true 25.7% | cycle6=visible 5.2%±0.0% / true 26.0%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 |
family_centroid_jsd_mean=0.161

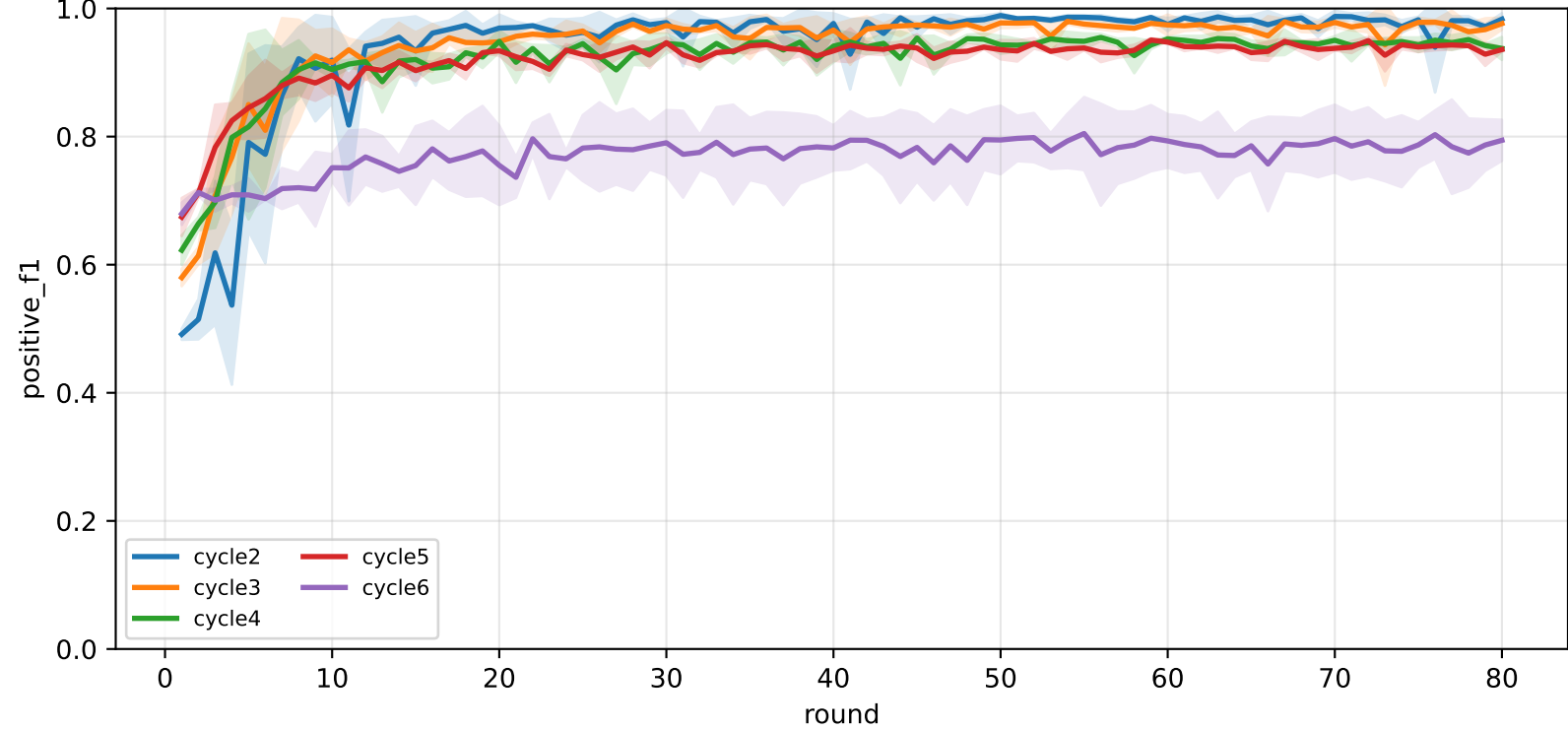
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	91.3% ± 1.0%	92.4% ± 0.7%	98.4% ± 0.5%	97.7% ± 0.5%	94.7% ± 1.2%	93.5% ± 0.9%	77.8% ± 3.3%
FedAvg	92.5% ± 0.4%	93.3% ± 0.3%	98.7% ± 0.3%	97.9% ± 0.5%	94.3% ± 0.2%	91.5% ± 1.5%	84.1% ± 0.9%
FedProx	92.9% ± 0.4%	93.8% ± 0.4%	98.8% ± 0.3%	98.2% ± 0.3%	95.4% ± 0.6%	92.5% ± 2.0%	83.8% ± 0.7%
GCFL+	92.6% ± 0.5%	93.4% ± 0.5%	98.9% ± 0.3%	98.0% ± 0.9%	94.0% ± 1.8%	93.2% ± 0.2%	82.8% ± 2.4%

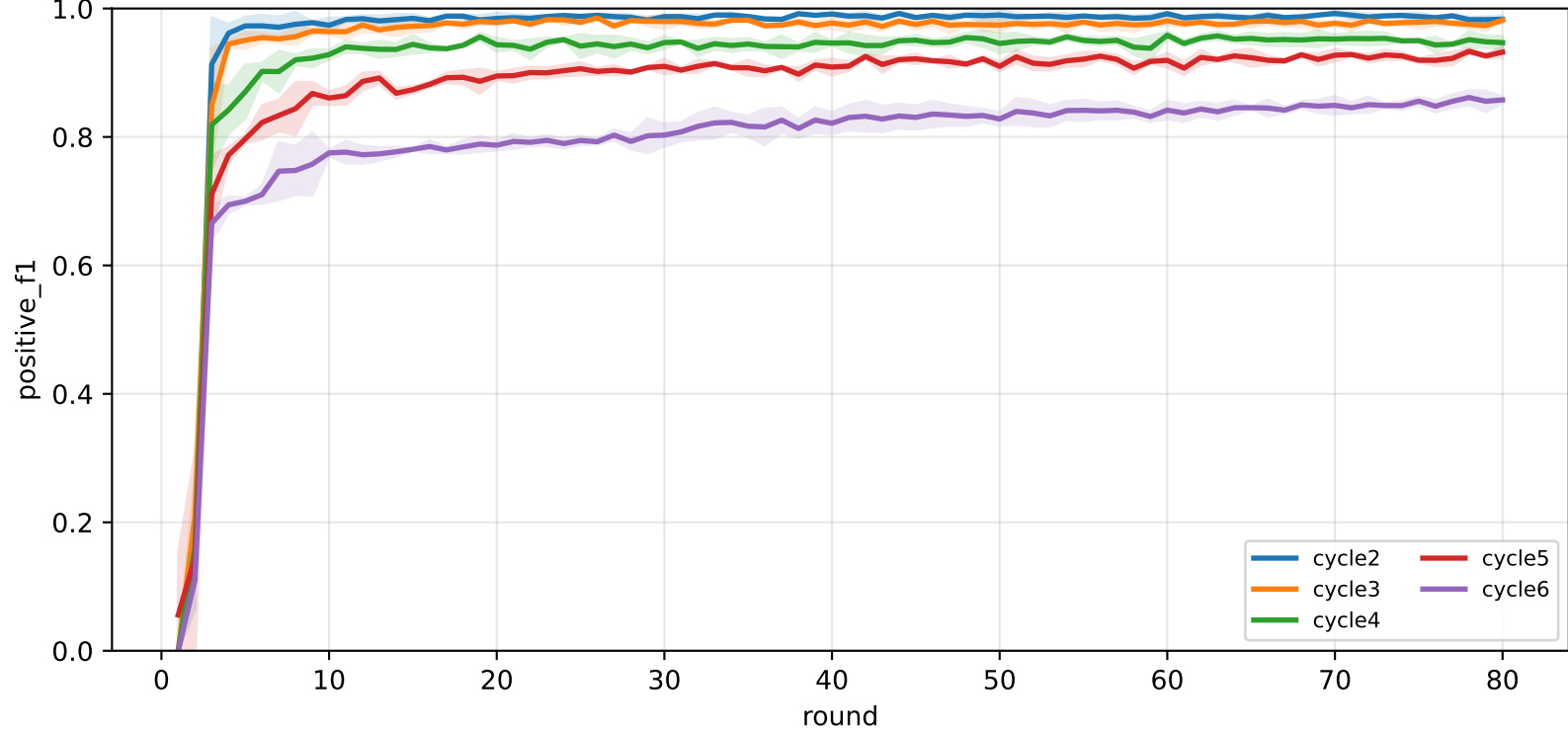
Validation loss comparison



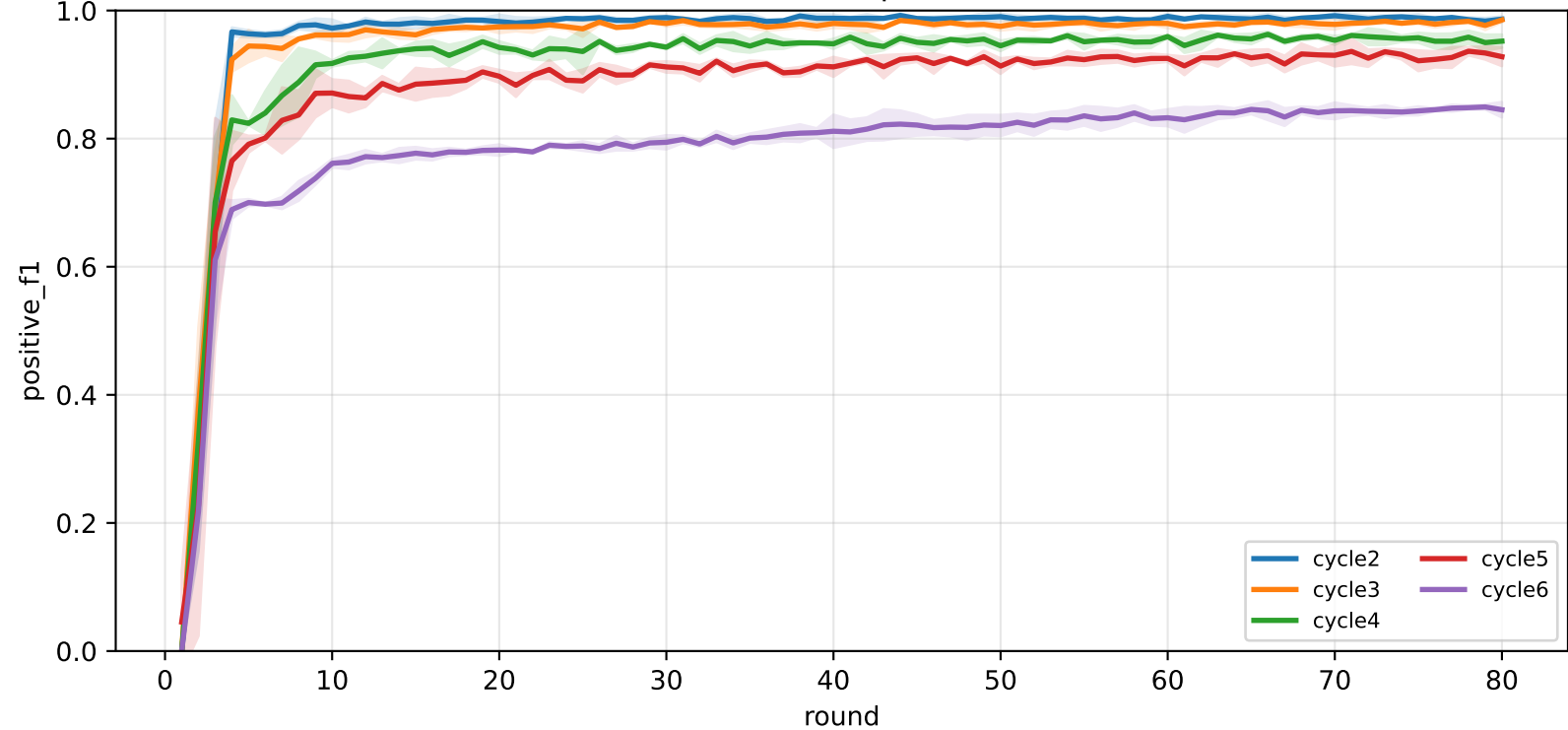
Pooled local centralized positive F1



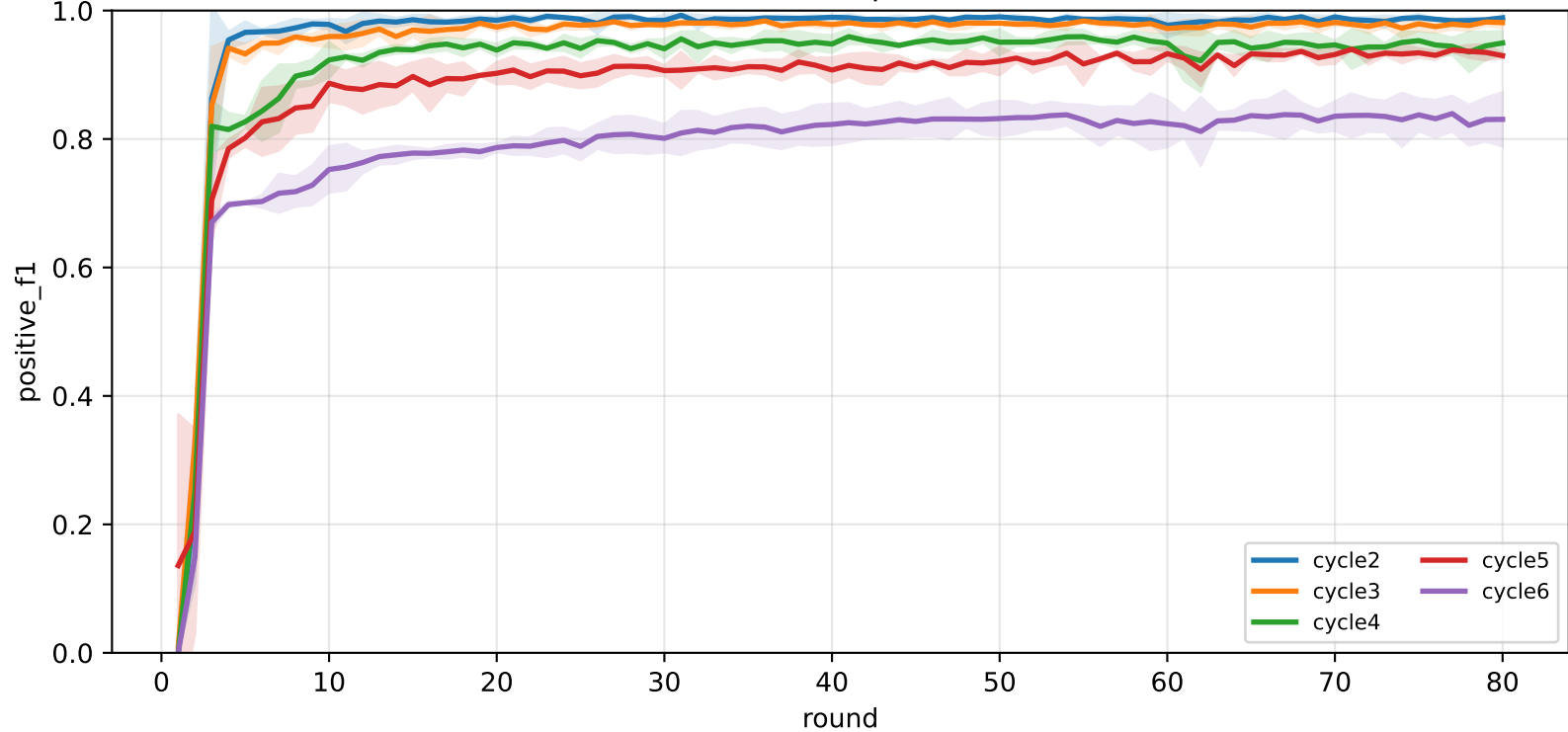
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1

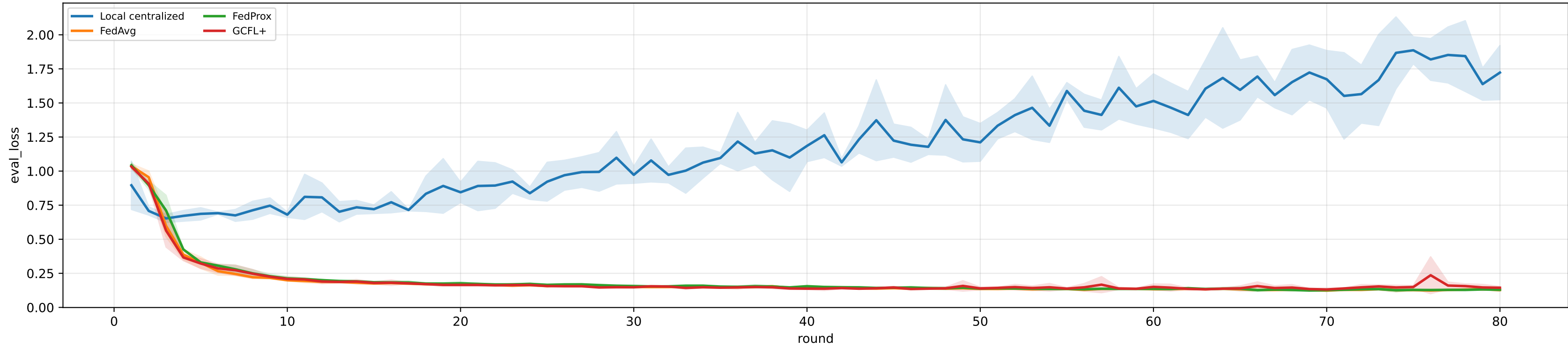


family=specialized_cycle6 | n_clients=1 | cycle2=visible 3.9%±0.0% / true 19.4% | cycle3=visible 4.7%±0.0% / true 23.6% | cycle4=visible 5.1%±0.0% / true 25.6% |
cycle5=visible 5.1%±0.0% / true 25.7% | cycle6=visible 26.0%±0.0% / true 26.0% |
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.161 | task_jsd_median=0.164 | task_jsd_max=0.168 |
family_centroid_jsd_mean=0.161

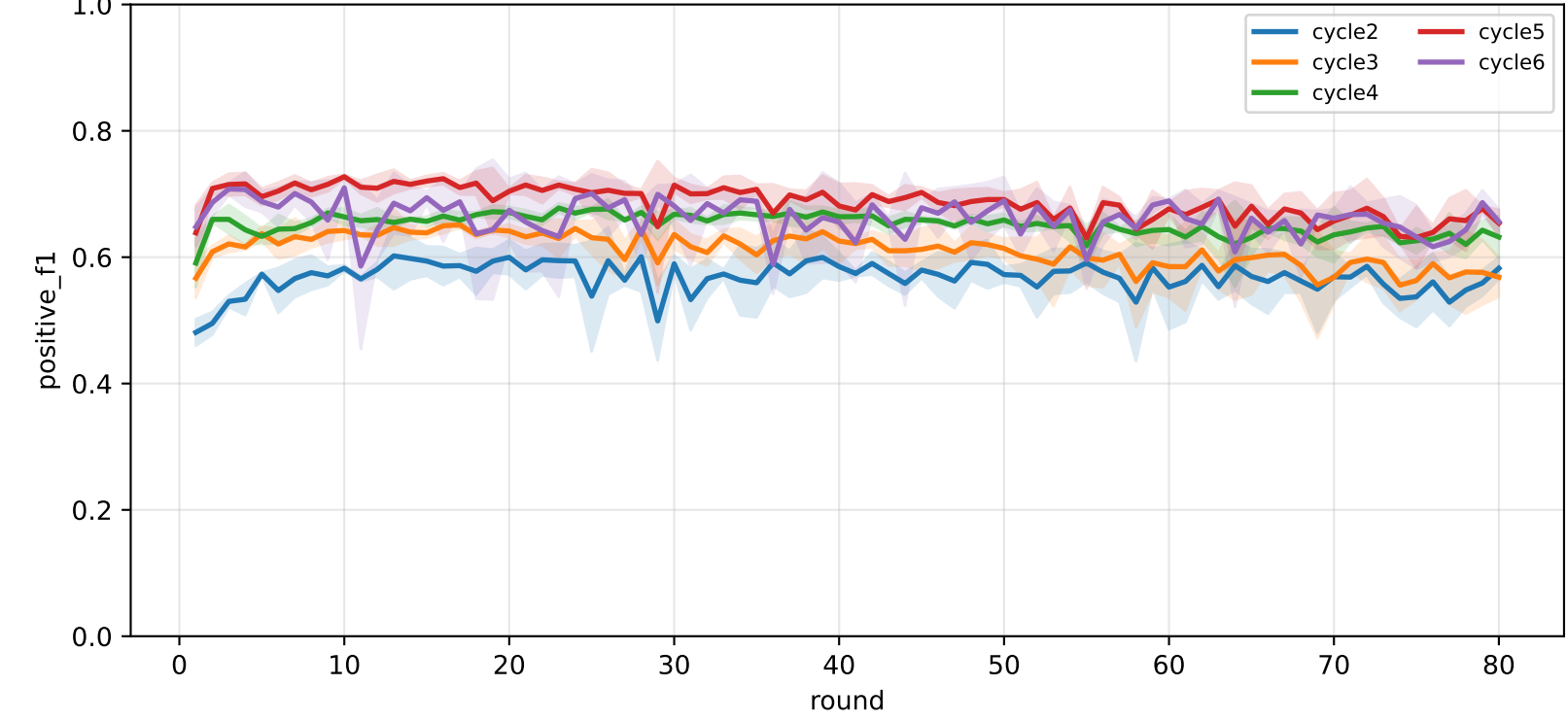
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	64.1% ± 1.1%	63.5% ± 1.0%	53.6% ± 1.8%	60.5% ± 0.7%	63.8% ± 0.9%	69.7% ± 0.8%	70.1% ± 1.9%
FedAvg	92.5% ± 0.2%	93.3% ± 0.1%	98.7% ± 0.2%	97.7% ± 0.7%	94.4% ± 0.9%	91.4% ± 1.5%	84.3% ± 1.1%
FedProx	92.8% ± 0.3%	93.7% ± 0.3%	98.8% ± 0.4%	98.0% ± 0.6%	95.3% ± 0.8%	92.4% ± 2.0%	83.6% ± 0.9%
GCFL+	92.0% ± 0.3%	92.9% ± 0.3%	98.3% ± 0.4%	97.4% ± 0.9%	94.4% ± 1.5%	90.7% ± 1.7%	83.5% ± 2.3%

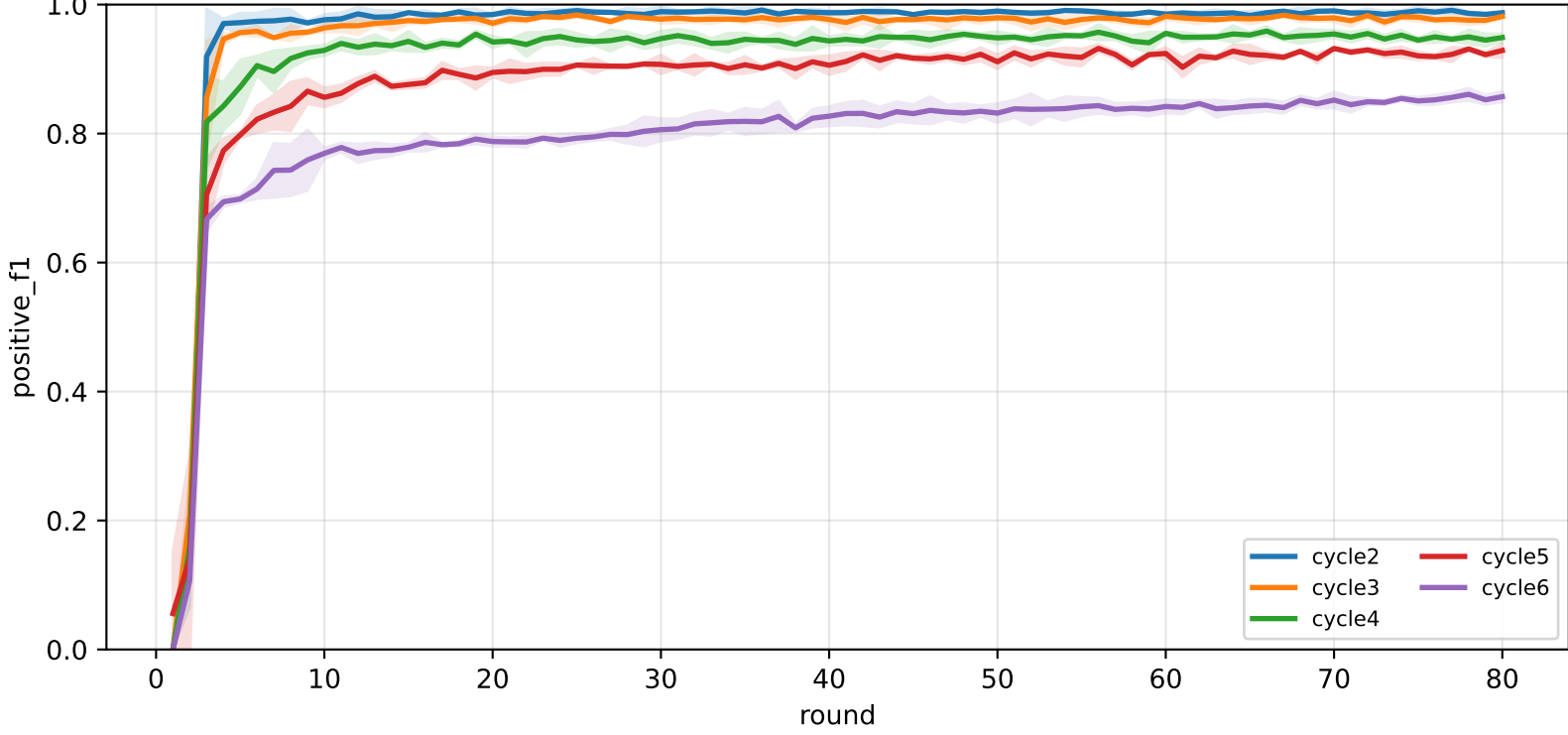
Validation loss comparison



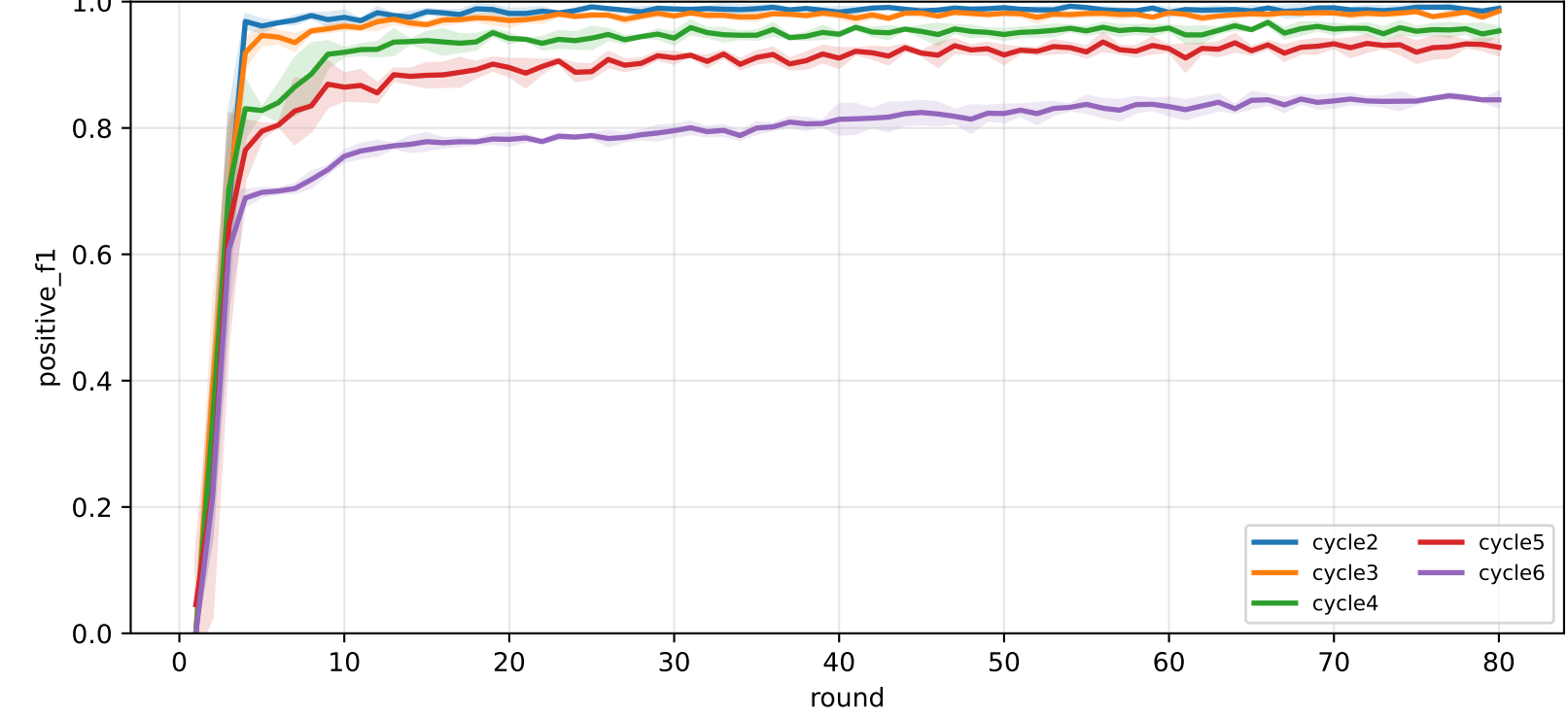
Pooled local centralized positive F1



Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1

